

Nicholas Kashani Motlagh

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Education

Ohio State University

Ph.D., Computer Science and Engineering

Columbus, OH

Conferred Aug 2026

Dissertation: *Answering Under Uncertainty: Abstention, Ambiguity, and Recoverability*

M.S., Computer Science and Engineering, May 2025 | B.S., CSE with Honors, May 2021

Experience

Computer Engineer II

May 2025 – Present

DCS Corporation / Air Force Research Laboratory, Dayton, OH

- Built evidence-grounded QA systems: small open models answer from retrieved documents, cite the supporting passage, and abstain when evidence falls short; LoRA on teacher-model synthetic data, paired base-vs-adapted evaluation.
- Built paired-outcome evaluation pipelines across 25,870 held-out QA questions, measuring when retrieval-based revision preserves, repairs, or harms an answer.
- Trained and evaluated LoRA models for answer/refine/abstain decisions (PyTorch, Hugging Face, Slurm).

Graduate Research Associate, Computer Vision Lab

Aug 2021 – Aug 2026

Ohio State University, Columbus, OH

- **Assessing the Role of Imagery in Multimodal Machine Translation** (WMT 2024): introduced ImageCoMMuTE metrics separating visual understanding from final translation decisions.
- **Naturally Constrained Reject Option Classification** (MVA 2025): developed Binomial-CDF reject-option constraints with per-class thresholds across synthetic, image, and text datasets.
- **Learning When to Say “I Don’t Know”** (ISVC 2022, Springer Best Paper Award): per-class softmax thresholds with no rejection cost; CIFAR-100 selective accuracy 88.3% → 97.8% at 77.3% coverage (public code).

Graduate Teaching Associate / Course Grader, Ohio State University

Aug 2023 – Dec 2025

- GTA through May 2025, then course grader through Dec 2025; machine learning and NLP coursework and Excel-based computing labs.

Graduate Research Intern

Summers 2022–2024

Air Force Research Laboratory, Dayton, OH

- Summer 2024: Adapted JEPA and MAE transformers for multimodal EO/SAR representation learning in low-data regimes.
- Summer 2023: Developed Reject Option Beam Search to improve machine translation quality at large beam widths.
- Summer 2022: Built the end-to-end training procedure for Naturally Constrained Reject Option Classification.

Undergraduate Research Intern, Air Force Research Laboratory

Summers 2020–2021

- **Temporal Satellite Imagery** (ICCV Workshop 2021): built an OpenStreetMap-guided Python pipeline for remote-sensing change detection and classification datasets.

Selected Publications and Manuscripts

- [1] **N. Kashani Motlagh** et al. “Measuring and Predicting Recoverability in Retrieval-Augmented Revision.” Revised after ACL Rolling Review; arXiv version in preparation, 2026.
- [2] **N. Kashani Motlagh** et al. “Naturally Constrained Reject Option Classification.” *Machine Vision and Applications* 36, Article 9, 2025.
- [3] **N. Kashani Motlagh** et al. “Assessing the Role of Imagery in Multimodal Machine Translation.” *WMT*, pp. 1428–1439, 2024.
- [4] **N. Kashani Motlagh** et al. “Learning When to Say ‘I Don’t Know.’” *ISVC*, pp. 196–210, 2022. Springer Best Paper Award.
- [5] **N. Kashani Motlagh** et al. “A Framework for Semi-automatic Collection of Temporal Satellite Imagery for Analysis of Dynamic Regions.” *ICCV Workshop*, pp. 704–712, 2021.

Technical Skills

Skills: LLM evaluation harnesses, selective prediction and abstention, calibration (ECE, Brier), RAG, LoRA; Python, PyTorch, Hugging Face, vLLM, FAISS, Git, Slurm, Singularity, LaTeX

Service: Reviewer for ICCV 2023, CVPR 2023, ECCV 2022, CVPR 2022; volunteer, HackOHI/O 2023